Power Converter Analysis and Design

ASSIGNMENT - 3

1. In comparing a single-phase diode bridge rectifier with a single-phase IGBT bridge rectifier, both operating at similar input voltage and power levels and operated to exploit their respective capabilities, which of the following statements are correct?
    A. The power loss in the IGBT bridge is likely to be larger.
    B. The input current THD in the IGBT bridge is likely to be lesser than that in the diode bridge.
    C. The output voltage in the diode bridge is likely to be controllable, while not so in the case of the IGBT bridge.
    D. The output voltage in the case of the IGBT bridge is likely to be controllable, while not so in the case of the diode bridge.

2. In comparing the behavior of a single-phase and a three-phase diode bridge rectifier, both equipped with the same value of LC filter on the DC side and with the inductor current assumed to be smooth, which of the following statements are correct?
    A. The single-phase diode bridge has a higher input current THD than the three-phase case.
    B. The three-phase bridge with its LC filter delivers a DC voltage with lesser ripple than the single-phase case with its respective LC filter.
    C. The frequency of ripple voltage at the DC terminals of the rectifier is the same in both cases.
    D. The three-phase bridge output voltage is controllable, while not so in single-phase bridges.

3. Why are fins used in a heat sink? Is it possible to have heat sinks without fins?
    A. Fins are used to increase the effective surface area for heat transfer, thereby enhancing cooling.
    B. Heat sinks without fins are possible, but their thermal performance is much lower compared to heat sinks with fins for the same mass and cooling condition.
    C. Fins are used to decrease the thermal conductivity of the material.
    D. Fins mainly work by increasing radiation heat transfer only.

4. Consider the circuit and phasor diagram shown in Figure 1. The circuit consists of a single phase H-bridge with the input inductance L, a DC link capacitor C_{bus}, and a bi-directional DC–DC converter connected to a battery of voltage V_{bat}. The phasor diagram illustrates the relationship among the source voltage v_s, source current i_s, inductor voltage v_L, and H-bridge mid-point voltage v_{ab}.

   Based on the phasor diagram, identify the operating mode of the input H-bridge.

   A. Rectification with unity power factor (UPF, AC to DC).

   B. Inversion with unity power factor (UPF, DC to AC).

   C. Rectification with lagging power factor.

   D. Inversion with leading power factor.

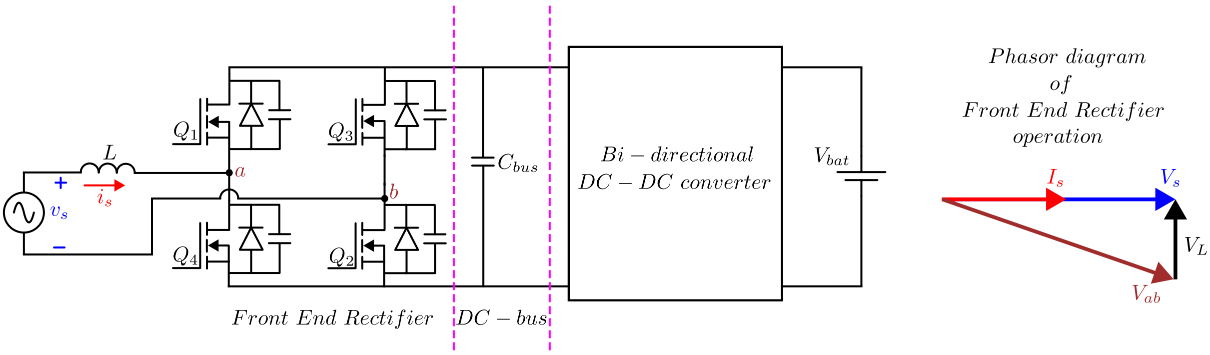

Figure 1: Front end rectifier with DC-DC converter connected to a battery

5. A MOSFET used in a step-down converter has an ON-state loss of 50 W and a switching loss given by $10^{-3} f_s$ (in watts), where f_s is the switching frequency in hertz. The junction-to-case thermal resistance $R_{\theta,jc}$ is $1\,°C/W$ and the maximum junction temperature $T_{j,max}$ is $150\,°C$. The MOSFET is mounted on a heat sink and the ambient temperature $T_a = 35\,°C$. If the switching frequency is 25 kHz, what is the maximum allowable value of case-to-ambient thermal resistance $R_{\theta,ca}$.

6. A MOSFET that is used in an inverter is dissipating 20 W. The junction to case thermal resistance is $R_{\theta,jc} = 0.5\,°C/W$ and case-to-heat sink thermal resistance is $R_{\theta,ch} = 1.5\,°C/W$. The heat sink used ensures that the device junction temperature does not exceed $100\,°C$. If the ambient temperature in the neighborhood of the heat sink is $50\,°C$, then estimate the device case temperature.

7. A capacitor filter rectifier is supplied from a 50 Hz AC mains source. The load power is 500 W, and the peak to peak output voltage ripple must not exceed 15 V.

   Determine the required filter capacitance for each of the following cases:

   1. Single-phase half-wave rectifier (supply: 230 V L–N)
   2. Single-phase full-wave rectifier (supply: 230 V L–N)
   3. Three-phase full-wave rectifier (supply: 400 V L–N)

   Assume that ideal diodes are used in all cases.

8. A three-phase full-bridge diode rectifier is supplied by a balanced 400 V, 50 Hz three-phase AC source. The load is 100 kW. Design a DC-side LC filter such that it provides 40 dB attenuation for the dominant non-DC frequency component. The inductor current ripple at this dominant frequency does not exceed 20% of the average load current. The normalized second order filter frequency response is shown in Fig. 2, where the attenuation (in dB) is plotted against the normalized frequency ratio ω/ω_n. Use this figure to determine the required filter natural frequency ω_n for designing the LC filter.

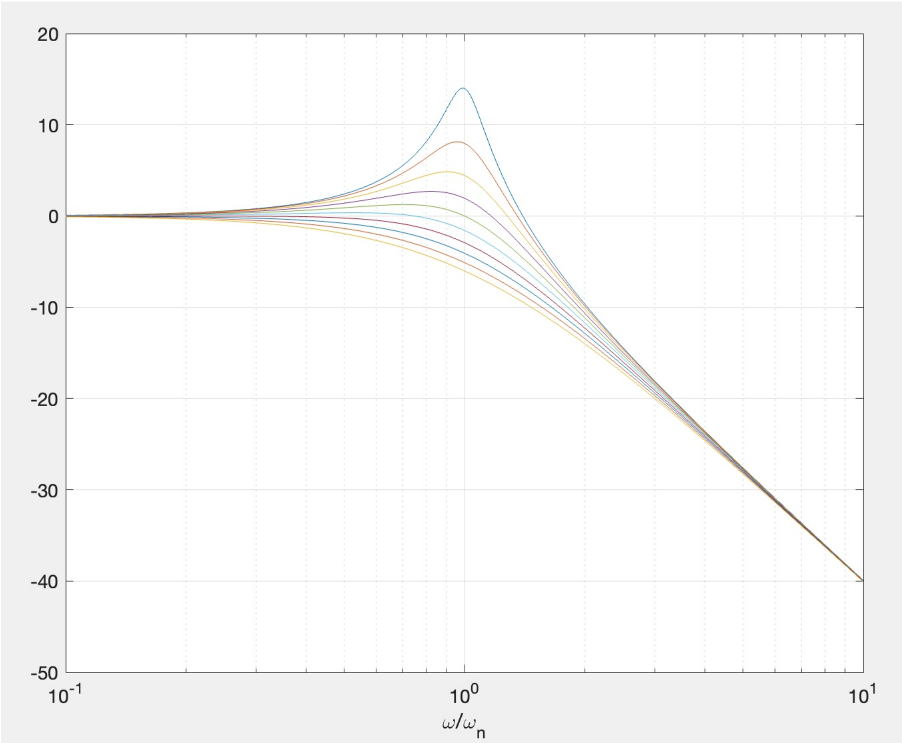

Figure 2: Normalized second order filter frequency response

9. In a single-phase Active Power Factor Correction (PFC) circuit, the AC supply is a sinusoidal voltage source of RMS voltage $V_s = 230\,\text{V}$ at a frequency of $f = 50\,\text{Hz}$. The circuit feeds a purely resistive load of resistance $R = 37\,\Omega$. The rated output power delivered to the load is specified as $P_{\text{load}} = 3.9\,\text{kW}$.

   Assume that the PFC controller shapes the input current to be sinusoidal and in phase with the input voltage (i.e., unity power factor operation). Neglect the effect of the 100 Hz ripple on the DC output voltage.

   Determine the following:

   1. The amplitude of the inductor current reference.
   2. The average load (DC) voltage.
   3. The load current.

10. Derive expressions to show that in a single-phase power factor controlled rectifier (PFC, having a diode bridge and boost converter in series), there is a dominant voltage ripple at the output capacitor at double the power supply frequency. What are the options to keep the ripple voltage at desirable low levels?